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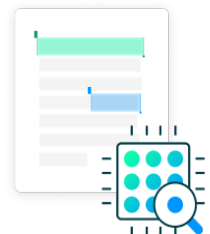
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# Title - “AI Driven Human Emotion Recognition Supporting Mental Health “

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Section : A  
Course : Research Methodology

## 1. Motivation :

My interest in the field of Computer Science has been around developing technologies that impact people's lives, felt rather than abstract. Many areas of computing are interesting but I am most interested in where Artificial Intelligence (AI) and human well-being overlap. The concept that we might be able to design systems that actually comprehend, interpret and react to human needs on an emotional level is not just a technical puzzle but also an inspiring chance to do good. That interest has grown and evolved to a particular focus on how AI can play a role in assisting mental health, an area I have come to believe represents one of the most pressing and ill-served needs of our society today.

My motivation for choosing the topic "AI-driven human emotion recognition supporting mental health" as I see worldwide mental health pandemic expanding. Many suffer in silence, without the resources to afford therapy, or because it remains stigmatized; services simply aren't available. Moreover, people generally have a hard time monitoring how they feel over time and being able to notice patterns or triggers affecting their mood is also an arduous task. I think technology can be vitally important as a first-line support and monitoring tool. By building more smart systems that can detect human emotion from, for example, text, voice or faces accessible and low cost tools could be developed to help people learn more about their own mental health and be encouraged to ask for help.

This research is not about replacing therapists or human connection. Instead, it is about augmentation and accessibility. An AI system can be made to work 24/7 offering the option of using it whenever its user chooses without judgment and allowing them to collect potentially invaluable data in an objective manner for both themselves and their healthcare professionals. I want to add to that by building more accurate, ethical and empathetic AI models. I would like to conduct research that is not just the classification of emotions but understanding the context around those emotions and building systems that can provide insights and timely interventions which will promote a journey towards mental health. This is the ideal combination of my technical specialties AI and my long-standing wish that I could work on something with profound human and social value.

## 2. Research Proposal:

### i. Introduction

Mental health issues are a leading cause of disability worldwide, Yet a significant gap remains between the number of people needing care and those who receive it (Vaidyam et al., 2019). Recent global events have only accelerated this crisis, emphasizing the need for innovative and scalable responses. In that sense AI has a disruptive potential. In other words, AI based human emotion recognition refers to the use of machine learning and deep learning models to recognize and classify human emotions from multimodal sources such as text (journals or social media), voice and face expressions (Ganesan et al., 2022).

A lot of the work we're doing in this space is meaningful because it can democratize mental health support. Mobile applications or web-based platforms that utilize AI to provide inexpensive, always-available preliminary screening and mood tracking can serve as useful tools. This can help overcome barriers for treatment like cost, stigma and geographical isolation. Furthermore, these systems may also provide an objective and data-driven understanding of a person's positive mood states over time, for which individuals often have difficulty providing accurate self-reporting (Gomes et al., 2024). For clinicians, this data can be invaluable, providing a more comprehensive picture of a patient's condition between therapy sessions and enabling more personalized treatment plan for the patients.

This proposed research have the potential to contribute usably to Computer Science by expanding how AI interprets human context. Most existing systems place emotions in general categories like "happy," "sad" or "angry." But let's face it, human feelings are nuanced in the extreme. The aim of this study is to build models that go beyond the naïve notion if there is no discussion, then it did not happen and understand more accurately the fine grain context around emotion expression By creating an ecosystem that not only detects emotions but also tracks their progression and potential triggers, we can move from reactive to proactive mental health support, ultimately creating a significant positive impact on society.

## ii. Literature Review

Previous work There is a large literature on the use of AI in mental health. Early work was on single modalities. For example, In the field of Artificial intelligence, Natural Language Processing has been extensively utilized to analyze text for depression and anxiety symptoms. Al Maruf et al.,(2017) designed models to classify mood by text data and showcased the possibility of tracking well-being information. Furthermore, speech patterns (e.g. tone, pitch and speed) has been proved to have potential value in the prediction of psychological stress (Cheng et al., 2023).

On another visual field of AI, Convolutional Neural Networks have been established as the architecture of choice for FER. Esan, A.,et al. (2024) highlighted the effectiveness of neural networks in detecting faces and classifying emotions from images, a technique that can be used for real-time monitoring. Now the field is turning toward multimodal approaches contriving the shared signal from these sources. Ganesan et al. (2022) have suggested an "emotional health ecosystem" that processes edge device data where context information and emotional expressiveness ratio were combined to get more comprehensive view, especially for women's health.

However, there are critical gaps and challenges in the existing literature. A major issue is the lack of contextual understanding. As pointed out by Park, H. J. et al., (2018), an AI could accurately classify a weeping face as "sad," but the trained model does not know that tears of sadness are different from tears of joy. This may result in substantial misclassification. In addition, many models are trained on lab-controlled data rather than with the disruptive real-world data collected from personal devices.

Ethical considerations are also a major point of discussion. The collection of sensitive biometric and personal data raises serious concerns about privacy, data protection, and potential misuse (Shimada, K. 2023; Cruz-Gonzalez,et al.,2025). There is also a risk of over-reliance on these technologies, potentially leading to a dehumanization of care if not implemented carefully (Graham, et al., 2019). Finally, the question of bias in AI models is critical; if training data is not diverse, the systems will perform poorly for underrepresented groups, potentially worsening health disparities (Graham et al., 2019).

Despite these many and such obstacles, the opportunity is monumentally vast. In both low- and middle-income countries, AI can be deployed for monitoring (and possibly for delivering basic treatment techniques) (Singh, S. 2023). The crucial point to develop technology, with a cautious approach about its ethical and practical limitations as strongly pursued in this project proposal.

### iii .Objective

**Main Objective:** To design, develop and evaluate an AI enabled context-aware system for human emotion recognition that can offer actionable insights and help in mental health monitoring.

**Sub-Objectives:**

1. To develop a emotion recognition model which effectively captures comprehensive features from text, audio and visual inputs
2. To design and implement a context-analysis module in which the designed one enhances emotion classification accuracy by taking users' specific situation and historical information into account.
3. To create a user-friendly dashboard that visualizes emotional trends through time and offers tailored mental health resources recommended from scientific study.
4. To evaluate the system's effectiveness, usability and ethical adherence via a pilot study with a limited number of users.

### iv. Research Question

**Main Research Question:** How is an AI system with context-aware, multimodal learning can be designed to accurately detect human emotions and offer actionable aid in mental health monitoring?

**Sub-Questions:**

1. How to efficiently integrate information of text, speech and facial expression so that there is advancement in emotion recognition performance over using single-modal features?
2. What contextual factors (e.g., occasion, geographic location, recent behaviors) are most important for emotion classification and how can be they integrated into the AI model?
3. What is the system's usability and usefulness experienced by users in terms of gaining insights into their mental well-being?

## 3. Proposed Research Methodology :

The research will be carried out in three phases, with methodological approaches tailored to each question, and designed to provide a thorough exploration.

**For Research Sub-Question 1:**

To answer how different data types can be fused, a quantitative and experimental methodology will be used. We will begin by collecting a multimodal dataset. Publicly available datasets containing aligned text, audio, and video of emotional expressions (like MELD or IEMOCAP) will be used. We will then design a neural network architecture. We will design a neural network then. This will require independent feature extraction branches: a text-based Transformer model (e.g., BERT ) to understand semantic meaning (Al Maruf et al.,2017), a CNN followed by Recurrent Neural Networks (RNNs) for audio to capture temporal speech patterns (Cheng et al., 2023), and a CNN (like VGG or ResNet) for facial feature extraction (Esan, A. et al., 2024). The key methodological step will be testing different fusion strategies early fusion (combining raw data), mid-level fusion (combining extracted features), and late fusion (combining model decisions) to determine which approach yields the highest emotion classification accuracy. The performance will be measured using standard metrics like precision, recall, and F1-score.

### For Research Sub-Question 2:

Mixed-method will be used to identify and incorporate the contextual factors. First, a qualitative literature review will be conducted in order to have a list of potential contextual factors proposed in psychology and human-computer interaction (Park, H. J. et al., 2018). An exploratory (small-scale) qualitative study will be conducted in which we explore through surveys and interviews with potential users of the system and mental health professionals what context cues are perceived to be most relevant.. Based on this, we will design a context-analysis module. Methodologically, this module will use a combination of rule-based logic (e.g., "if emotion is 'anxious' and context is 'before a meeting', flag as 'situational anxiety'") and machine learning. We will train a separate model to predict the most likely context from available data (e.g., calendar entries, time, passively collected smartphone data with user permission) and use its output to weight the predictions of the multimodal emotion model. The improvement in accuracy from adding this context module will be quantitatively evaluated against the baseline multimodal model.

### For Research Sub-Question 3:

A qualitative and user-centric methodology will be used to assess the user experience. We will develop a functional prototype of the system with its visualization dashboard. A pilot study will then be conducted with a group of 20-30 participants over a one-month period. The methodology will involve collecting two types of data: (1) Usage data: Logs of how often users interact with the app and which features they use. (2) Qualitative feedback: This is the core of answering this question. We will carry out semi-structured interviews and use standardized usability questionnaires, such as the System Usability Scale (SUS), in the final pilot. 23 ) We will analyse the transcripts of interviews thematically to identify patterns of responses to questions about perceived usefulness, trust in the system, clarity of insights and overall satisfaction. This user feedback will be invaluable in iterating and refining the final design.

## 4. Ethical Considerations:

In a sensitive research area like this, protecting participants is the highest priority. We will have strict procedures to ensure confidentiality, anonymity and data safety of the project characterized in similar studies Shimada, K. (2023). and Cruz-Gonzalez et al. (2025).

**Data Privacy:** The collected data from the users will be of sensitive nature, whether its text, audio or video. We will protect them through a variety of means. First, all of the data will be encrypted both when it is being transmitted from the user's device to our servers and when it is stored ('encryption at rest and in transit'). Second, we will use data minimization only using the data we absolutely need to have in order for everything to work. For instance, raw video files can be processed by a user's or player device to extract only the relevant facial feature codes and then deleted without further retention of original video. In addition, we make use of secure servers and apply a robust access protocol and the data is retained only for a reasonably short promised period before it is irrecoverably erased.

**Confidentiality:** The principle of confidentiality is concerned with who can view the information that reaches us from external sources. To protect the confidentiality of participants, access to the de-identified data will be limited to a handful of named project researchers. Ethics training will be mandatory for all members of teams. Any data that is analyzed will be de-identified (pseudonymized) and made impossible to directly trace back to the originator. We will also have a defined data management plan stating who the data belongs to and when we can access it.

**Anonymity:** Anonymity is one step beyond confidentiality with respect to data protection, it means that even the researchers cannot link the data to the individual. In our study, we will ensure anonymity during the publication and reporting phase. When we present our findings or write papers, we will only use aggregated and anonymized data (e.g., "70% of users found the trend graph useful"). No individual's data or identifying information will ever be published. In the rare case that a direct quote from an interview is used, it will be thoroughly sanitized to remove any

potentially identifying details, and it will be referred to only by a randomly generated participant ID (e.g., "Participant P12"). By implementing these layered ethical safeguards, we aim to build a foundation of trust with our users, which is essential for any technology that deals with mental health.

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